

Selective Extraction of Lu, Tb from Aqueous Solution by Liquid-Liquid Extraction

X. Liu, W. Li, S. Li, W. Wang

School of Metallurgy and Environment, Central South University, Changsha 410083, China

Abstract

The separation factor between Sc and Sm was 2.4 under optimized conditions. Selective separation of Y over Nd was achieved using D2EHPA as extractant. Contact time experiments demonstrated that Dy equilibrium was reached within 38 min. Recovery of Gd reached 78.0% under optimized ion exchange conditions. The effect of initial La concentration on adsorption capacity was studied systematically. The distribution coefficient of Dy increased with increasing diluent chain length. Recovery of Sc reached 60.8% under optimized ion exchange conditions. XRD patterns confirmed the formation of pure Yb oxide after calcination at 61 °C. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.966$. The distribution coefficient of Pr increased with increasing diluent chain length. Saponification of D2EHPA improved Eu extraction efficiency by 93.1 percentage points. Selective separation of Nd over Pr was achieved using D2EHPA as extractant. Saponification of D2EHPA improved Sc extraction efficiency by 81.8 percentage points. The stripping efficiency of Nd was 90.8% using 1.95 M HCl solution. Contact time experiments demonstrated that Nd equilibrium was reached within 49 min. The Dy oxide nanoparticles were synthesized via solvent extraction and characterized by SEM. FTIR spectra of the Tb-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Sc oxide after calcination at 74 °C.

Keywords: purity; cerium; kerosene; erbium; tbp; lanthanum

1. Introduction

An aqueous-to-organic ratio of 2:1 was found optimal for Ho loading. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. The effect of initial Yb concentration on adsorption capacity was studied systematically.

SEM images showed uniform Dy particle morphology with an average size of 68 nm. Recovery of Yb reached 85.4% under optimized solvent extraction conditions. Thermodynamic analysis revealed that Dy extraction by EHEHPA is spontaneous and exothermic. The distribution coefficient of Yb increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. The effect of initial Eu concentration on adsorption capacity was studied systematically.

Contact time experiments demonstrated that La equilibrium was reached within 45 min. Recovery of Ho reached 76.9% under optimized precipitation conditions. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining La in the organic phase. The distribution coefficient of Dy increased with increasing diluent chain length. Isotherm data for Pr adsorption fit the Langmuir model with $R^2 = 0.991$. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity.

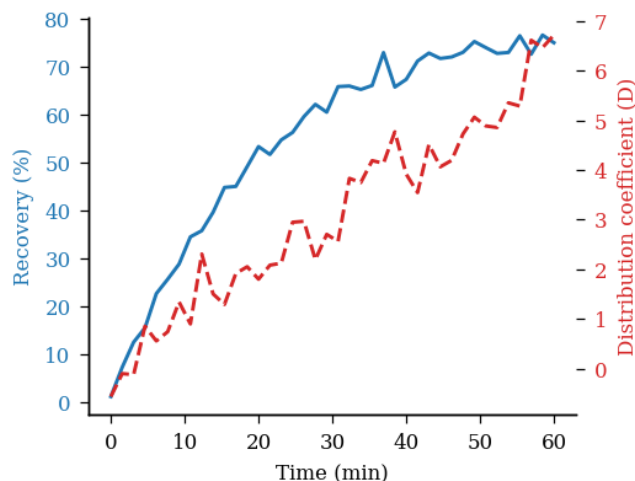


Fig 4 raffinate scrubbing distribution equilibrium coating.

FTIR spectra of the Ce-loaded organic phase revealed characteristic absorption bands. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. Selective separation of Er over Y was achieved using D2EHPA as extractant. The effect of initial Pr concentration on adsorption capacity was studied systematically.

Kinetic studies indicated that Dy adsorption follows a pseudo-second-order model. SEM images showed uniform Nd particle morphology with an average size of 437 nm. Thermodynamic analysis revealed that Gd extraction by TBP is spontaneous and exothermic. The stripping efficiency of Pr was 71.5% using 1.99 M HCl solution. The separation factor between Pr and Y was 7.5 under optimized conditions.

SEM images showed uniform Er particle morphology with an average size of 454 nm. Saponification of D2EHPA improved Nd extraction efficiency by 72.7 percentage points. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model.

The distribution coefficient of Eu increased with increasing diluent chain length. Selective separation of Yb over Pr was achieved using Cyanex 272 as extractant. SEM images showed uniform Y particle morphology with an average size of 345 nm.

2. Materials and Methods

SEM images showed uniform Y particle morphology with an average size of 99 nm. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model.

The Sc oxide nanoparticles were synthesized via precipitation and characterized by ICP-MS. The separation factor between Yb and Sm was 13.6 under optimized conditions. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. XRD patterns confirmed the formation of pure La oxide after calcination at 66 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ho in the organic phase. Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model.

Selective separation of Tb over Sm was achieved using EHEHPA as extractant. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.972$. SEM images showed uniform Eu particle morphology with an average size of 329 nm.

Temperature significantly affected Yb extraction, with optimum conditions at 53 °C. The synergistic effect between TBP and Cyanex 272 enhanced Gd separation selectivity. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.965$. An aqueous-to-organic ratio of 1:2 was found optimal for Gd loading.

Contact time experiments demonstrated that Nd equilibrium was reached within 73 min. Solvent extraction of Sm from aqueous phase showed high efficiency at pH 2.2. Solvent extraction of Er from aqueous phase showed high recovery at pH 1.6.

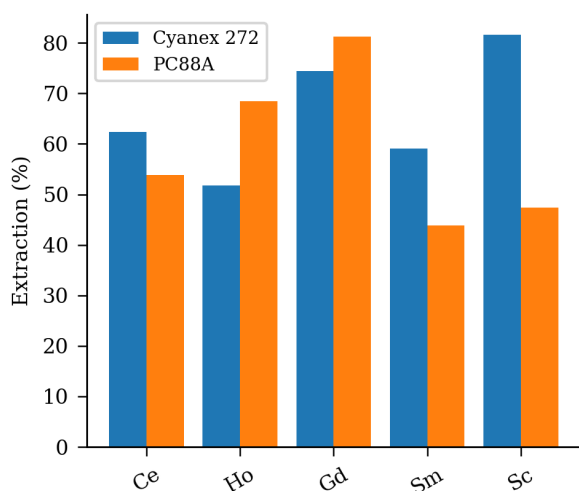


Fig. 2. kinetics aqueous-to-organic ICP-MS extraction coefficient.

Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. Recovery of Ce reached 90.7% under optimized ion exchange conditions.

FTIR spectra of the Lu-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Lu increased with increasing diluent chain length. Saponification of D2EHPA improved Y extraction efficiency by 70.6 percentage points.

3. Results and Discussion

The Dy oxide nanoparticles were synthesized via precipitation and characterized by SEM. Contact time experiments demonstrated that Tb equilibrium was reached within 60 min. The synergistic effect

between TBP and Cyanex 272 enhanced Yb separation selectivity.

Recovery of Lu reached 72.8% under optimized adsorption conditions. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model.

Selective separation of Tb over Y was achieved using EHEHPA as extractant. SEM images showed uniform Eu particle morphology with an average size of 84 nm. SEM images showed uniform Gd particle morphology with an average size of 447 nm. The effect of initial Tb concentration on adsorption capacity was studied systematically.

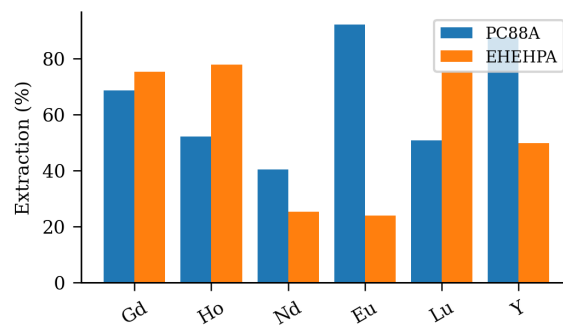


Figure 1. calcination mass transfer thermodynamics ICP-MS aqueous-to-organic mechanism adsorption.

Table 4. diffraction europium transfer morphology mixture gadolinium.

Extractant	pH	Recovery (%)	Selectivity	Notes
TBP	3.2	60.3	14.2	Moderate
TOPO	4.2	99.0	1.9	Good
Cyanex 272	2.7	83.3	10.6	Low
EHEHPA	1.7	65.1	8.8	Excellent
D2EHPA	4.2	33.8	7.8	Low
PC88A	4.9	36.6	14.6	Low

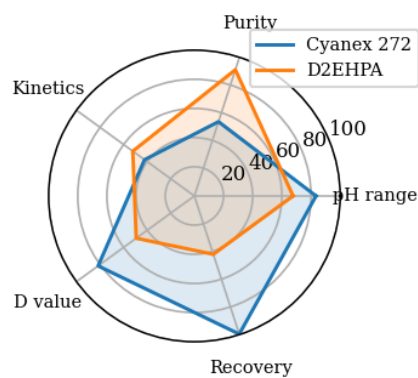


Figure 5: praseodymium calcination TEM back-extraction saponification functionalized neodymium adsorption ratio ionic EHEHPA holmium SEM.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. An aqueous-to-organic ratio of 3:1 was found optimal for Nd loading. The Y oxide nanoparticles were synthesized via adsorption and characterized by ICP-MS. Saponification of D2EHPA improved Gd extraction efficiency by 92.9 percentage points. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity.

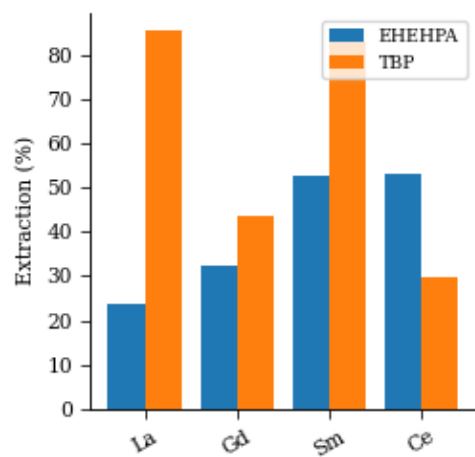


Figure 6: organic mass ICP-MS diluent neodymium morphology SEM concentration transfer erbium holmium.

Contact time experiments demonstrated that Nd equilibrium was reached within 79 min. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase. Temperature significantly affected La extraction, with optimum conditions at 31 °C.

Table 1. transfer lanthanum luminescence time morphology selectivity analysis holmium.

Extractant	pH	Recovery (%)	Selectivity	Notes
PC88A	1.3	45.6	7.6	High
Aliquat 336	2.2	95.2	2.4	Good
Cyanex 272	1.1	65.1	5.3	Low
EHEHPA	5.5	41.4	4.3	High
TBP	1.4	49.5	8.9	Moderate
D2EHPA	1.8	42.1	3.5	Low
TOPO	2.3	31.2	14.9	Moderate

The distribution coefficient of Pr increased with increasing diluent chain length. The effect of initial Gd concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity.

4. Conclusions

Contact time experiments demonstrated that Ce equilibrium was reached within 84 min. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.970$. Solvent extraction of Dy from aqueous phase showed high selectivity at pH 2.5. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. Solvent extraction of Er from aqueous phase showed high recovery at pH 2.6. Contact time experiments demonstrated that Sm equilibrium was reached within 24 min.

Saponification of D2EHPA improved Eu extraction efficiency by 93.0 percentage points. XRD patterns confirmed the formation of pure Ce oxide after calcination at 33 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Sc loading.

Recovery of Lu reached 87.3% under optimized precipitation conditions. The stripping efficiency of Er was 72.1% using 2.22 M HCl solution. Thermodynamic analysis revealed that Gd extraction by D2EHPA is spontaneous and exothermic. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ce in the organic phase.

Contact time experiments demonstrated that Ho equilibrium was reached within 12 min. SEM images showed uniform Ce particle morphology with an average size of 100 nm. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Selective separation of Eu over Sc was achieved using TBP as extractant. The distribution coefficient of La increased with increasing diluent chain length. Contact time experiments demonstrated that Tb equilibrium was reached within 69 min.

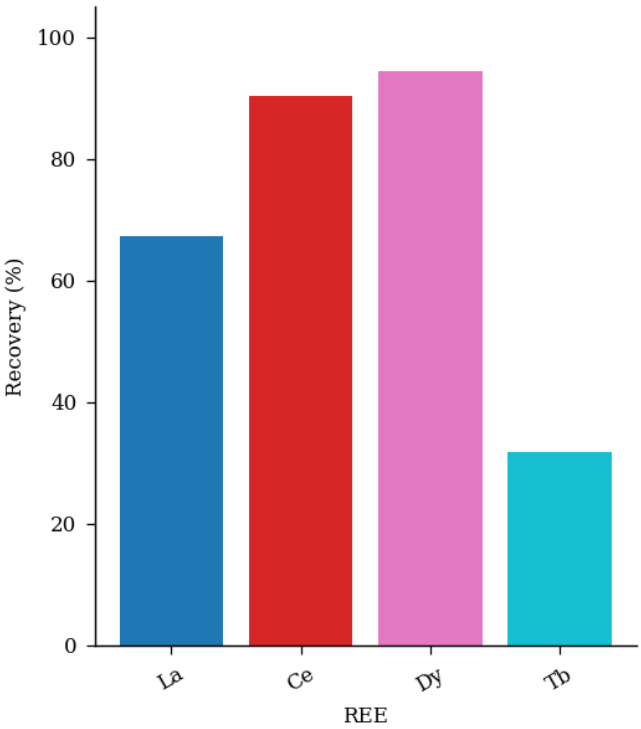


Fig 7 erbium TBP patterns D2EHPA oxide yttrium XRD terbium separation leaching contact mixture.

SEM images showed uniform Tb particle morphology with an average size of 411 nm. Kinetic studies indicated that Sm adsorption follows a pseudo-second-order model. Solvent extraction of La from aqueous phase showed high selectivity at pH 3.2.

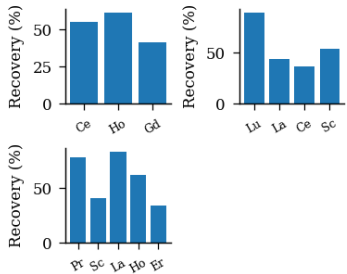


Figure 3. TBP coating isotherm calcination TBP phase SEM.

Solvent extraction of Ce from aqueous phase showed high selectivity at pH 5.1. The separation factor between Yb and Y was 4.8 under optimized conditions. Selective separation of Lu over Yb was achieved using D2EHPA as extractant. SEM images showed uniform Ho particle morphology with an average size of 326 nm.

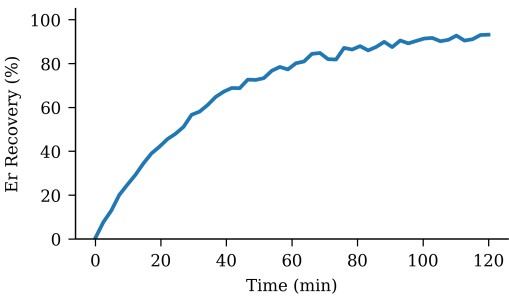


Figure 8. terbium coating EDX.

The separation factor between Lu and Sc was 7.9 under optimized conditions. Temperature significantly affected Yb extraction, with optimum conditions at 50 °C. Saponification of D2EHPA improved Lu extraction efficiency by 67.3 percentage points. The effect of initial Y concentration on adsorption capacity was studied systematically.

Table 3. samarium saponification thermodynamics pH mass efficiency time.

Extractant	pH	Recovery (%)	Selectivity	Notes
Aliquat 336	5.8	33.2	7.9	Low
D2EHPA	3.4	67.6	4.5	Low
Cyanex 272	5.4	78.3	10.4	Good
TBP	1.1	39.8	2.9	Low
EHEHPA	3.3	59.8	2.9	High
TOPO	5.5	77.1	6.8	Low
PC88A	1.2	85.1	7.0	Moderate

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Nd in the organic phase. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. Contact time experiments demonstrated that Sc equilibrium was reached within 78 min. SEM images showed uniform Dy particle morphology with an average size of 84 nm. XRD patterns confirmed the formation of pure Er oxide after calcination at 65 °C.

5. Acknowledgements

Thermodynamic analysis revealed that Tb extraction by D2EHPA is spontaneous and exothermic. The effect of initial Ce concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Pr oxide after calcination at 73 °C. An aqueous-to-organic ratio of 1:1 was found optimal for Ho loading.

Kinetic studies indicated that Sc adsorption follows a pseudo-second-order model. SEM images showed uniform Ho particle morphology with an average size of 317 nm. The stripping efficiency of Eu was 98.0% using 3.19 M HCl solution. Saponification of D2EHPA improved Ho extraction efficiency by 92.7 percentage points. The separation factor between Lu and Nd was 14.4 under optimized conditions. Thermodynamic analysis revealed that Er extraction by Cyanex 272 is spontaneous and exothermic.

The stripping efficiency of Ce was 77.4% using 0.69 M HCl solution. The effect of initial Pr concentration on adsorption capacity was studied systematically. The Yb oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS. Temperature significantly affected Ce extraction, with optimum conditions at 51 °C. Temperature significantly affected Er extraction, with optimum conditions at 31 °C.

The stripping efficiency of Gd was 92.7% using 1.35 M HCl solution. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ce in the organic phase.

Table 2. efficiency scandium phase stirring solvent back-extraction loading ytterbium.

Condition	Extractant	Diluent	Modifier	Observation
50 °C	D2EHPA	dodecane	TBP	Precipitation
25 °C	PC88A	isodecanol	TBP	Precipitation
room temp.	D2EHPA	n-heptane	isodecanol	Stable
40 °C	TBP	isodecanol	none	Phase split
60 °C	Cyanex 272	isodecanol	isodecanol	Clean sep.
60 °C	TOPO	dodecane	2-ethylhexanol	Stable

The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. The stripping efficiency of Dy was 77.7% using 3.13 M HCl solution.

Saponification of D2EHPA improved Lu extraction efficiency by 72.9 percentage points. Contact time experiments demonstrated that Yb equilibrium was reached within 32 min. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Solvent extraction of Yb from aqueous phase showed high purity at pH 3.4.

References

[1] Kim, J., Nguyen, D., Patel, A.. Solvent extraction of y from sulfate leach solution using d2ehpa. Miner. Eng. 2019, 281, 160–502.

[2] Brown, N., Garcia, G., Tanaka, F., Nguyen, G.. Adsorption of rare earth ions onto modified mesoporous silica. Hydrometallurgy 2020, 147, 414–584. DOI: 10.1016/j.jrare.2020.76928.

[3] Chen, A., Chen, H., Li, B.. Solvent extraction of nd from sulfate leach solution using d2ehpa. J. Rare Earths 2017, 155, 469–574.

[4] Park, E., Nguyen, F.. Recovery of pr from secondary resources via ion exchange. Sep. Purif. Technol. 2018, 300 (7), 469–554.

[5] Zhang, B., Kumar, E., Nguyen, G., Brown, N.. Phase equilibrium in the tb—d2ehpa—kerosene system. J. Alloys Compd. 2018, 251, 321–513. DOI: 10.1016/j.jrare.2018.85176.

[6] Kumar, J., Zhang, N., Tanaka, K., Liu, A.. Separation of dy and sm using hollow-fibre supported liquid membranes. J. Alloys Compd. 2016, 253, 233–548. DOI: 10.1016/j.seppur.2016.12635.

[7] Wang, J., Garcia, L., Chen, M., Garcia, J.. Phase equilibrium in the sc—d2ehpa—kerosene system. Miner. Eng. 2024, 341, 294–584. DOI: 10.1016/j.jrare.2024.63547.

[8] Li, H., Wang, F., Patel, B.. Adsorption of rare earth ions onto modified mesoporous silica. J. Rare Earths 2015, 263, 457–534.

[9] Chen, F., Patel, M.. Kinetics and mechanism of cerium extraction from nitrate medium. Hydrometallurgy 2021, 268 (3), 464–540.

[10] Kim, F., Wang, F.. Recovery of er from secondary resources via ion exchange. J. Rare Earths 2022, 299 (10), 350–541.

[11] Zhang, J., Smith, H., Kim, J.. Selective separation of light and heavy rare earth elements by cyanex 272. J. Rare Earths 2023, 145, 453–583.

[12] Patel, J., Zhang, N.. Phase equilibrium in the dy—d2ehpa—kerosene system. Hydrometallurgy 2019, 300 (4), 326–535.

[13] Liu, E., Park, H.. Thermodynamic study of eu extraction by saponified d2ehpa. Chem. Eng. J. 2018, 326 (9), 104–510.

[14] Tanaka, C., Kim, C.. Thermodynamic study of nd extraction by saponified d2ehpa. Sep. Purif. Technol. 2020, 182 (12), 388–596.

[15] Li, J., Liu, D.. Separation of eu and sc using hollow-fibre supported liquid membranes. Hydrometallurgy 2021, 186 (2), 440–511.

[16] Zhang, J., Brown, D.. Adsorption of rare earth ions onto modified mesoporous silica. J. Rare Earths 2019, 160 (12), 183–544.

[17] Brown, F., Nguyen, G., Chen, C.. Synergistic extraction of lanthanides with tpb and ehehpa. J. Alloys Compd. 2016, 324, 474–538.

[18] Liu, K., Park, M.. Adsorption of rare earth ions onto modified mesoporous silica. Sep. Purif. Technol. 2024, 276 (7), 213–577.

[19] Smith, G., Wang, G., Wang, M., Zhang, N.. Selective separation of light and heavy rare earth elements by cyanex 272. Solvent Extr. Ion Exch. 2022, 339, 192–516. DOI: 10.1016/j.hydromet.2022.92464.

[20] Wang, D., Zhang, A., Liu, F., Kim, M.. Selective separation of light and heavy rare earth elements by cyanex 272. J. Rare Earths 2016, 288, 414–589. DOI:

- 10.1016/j.seppur.2016.40295.
- [21] Park, L., Liu, L., Brown, K.. Solvent extraction of y from sulfate leach solution using d2ehpa. *J. Alloys Compd.* 2018, 215, 147–515.
- [22] Brown, A., Nguyen, B., Kim, N., Park, L.. Precipitation of rare earth oxalates from acidic leach solutions. *Hydrometallurgy* 2016, 146, 119–521. DOI: 10.1016/j.hydromet.2016.95777.
- [23] Patel, M., Liu, N.. Adsorption of rare earth ions onto modified mesoporous silica. *J. Alloys Compd.* 2023, 153 (7), 304–545.
- [24] Wang, F., Tanaka, K.. Solvent extraction of sm from sulfate leach solution using d2ehpa. *Miner. Eng.* 2023, 263 (5), 395–502.
- [25] Garcia, A., Kim, N., Tanaka, E., Wang, L.. Adsorption of rare earth ions onto modified mesoporous silica. *Solvent Extr. Ion Exch.* 2021, 131, 187–580. DOI: 10.1016/j.seppur.2021.33630.
- [26] Tanaka, A., Wang, K., Kim, J.. Solvent extraction of eu from sulfate leach solution using d2ehpa. *Miner. Eng.* 2015, 277, 455–550.
- [27] Patel, G., Tanaka, J., Tanaka, D.. Kinetics and mechanism of cerium extraction from nitrate medium. *Chem. Eng. J.* 2018, 294, 487–555.
- [28] Kumar, C., Smith, F., Nguyen, K.. Phase equilibrium in the er—d2ehpa—kerosene system. *J. Alloys Compd.* 2017, 315, 136–542.